Lec 11. Likelihood Ratio Tests

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Learning objectives

After this section, you should be able to ...

- understand the motivation of likelihood ratio (LR) test
- understand the connections between LR test, Wald test and Score test
- find the LR tests for specific problems

11.1 Likelihood Ratio Test

In this section we introduce intuitive and efficient procedures that can be used when no optimal methods are available and that are natural for multi-dimensional parameters.

Suppose the sample $\mathbf{X} = (X_1, \ldots, X_n)$ has pdf/pmf $f(\mathbf{x}, \theta)$, and consider

$$H_0 : \theta \in \Theta_0 \leftrightarrow H_1 : \theta \in \Theta_1$$

where $\Theta = \Theta_0 \cup \Theta_1$ is a subset of $\mathbb{R}^k$.

- For $\Theta_0 = \{\theta_0\}$ and $\Theta_1 = \{\theta_1\}$, the optimal test statistic is

$$\frac{f(\mathbf{X}, \theta_1)}{f(\mathbf{X}, \theta_0)}$$

- If we have composite hypotheses, we could instead use

$$\tilde{\Lambda}_n = \frac{\sup_{\theta \in \Theta_1} f(\mathbf{X}, \theta)}{\sup_{\theta \in \Theta_0} f(\mathbf{X}, \theta)}.$$

Since the critical value will be larger than 1, it will not change the test if we replace this statistic by $\tilde{\Lambda}_n \vee 1$.

Definition

Denote the likelihood ratio (LR)

$$\Lambda_n(\mathbf{x}) = \left(\tilde{\Lambda}_n \vee 1\right) = \frac{\left(\sup_{\theta\in\Theta_1} f(\mathbf{x},\theta)\right) \vee \left(\sup_{\theta\in\Theta_0} f(\mathbf{x},\theta)\right)}{\sup_{\theta\in\Theta_0} f(\mathbf{x},\theta)}$$

$$= \frac{\sup_{\theta\in\Theta_0\cup\Theta_1} f(\mathbf{x},\theta)}{\sup_{\theta\in\Theta_0} f(\mathbf{x},\theta)} = \frac{L(\widehat{\theta}_n)}{L(\widehat{\theta}_n^R)}$$

Then a level α likelihood ratio test (LRT) is given by

$$\phi(\mathbf{x}) = \begin{cases} 1, & \Lambda_n(\mathbf{x}) > c \\ r, & \Lambda_n(\mathbf{x}) = c \\ 0, & \Lambda_n(\mathbf{x}) < c \end{cases}$$

with c, r satisfying $\sup_{\theta\in\Theta_0} E_\theta \phi(\mathbf{X}) = \alpha$.

11.2 Find the LRT

General procedure to find the LRT of $H_0 : \theta \in \Theta_0 \leftrightarrow H_1 : \theta \in \Theta_1$:

- Determine Θ_0 and Θ, the likelihood function.
- Calculate the unconstrained MLE $\widehat{\theta}_n$ of θ.
- Calculate the constrained MLE $\widehat{\theta}_n^R$ of θ under Θ_0.
- Form $\Lambda_n(\mathbf{x})$
- **Find a function h** that is strictly increasing on the range of Λ_n such that $h(\Lambda_n(\mathbf{X}))$ has a simple form and a tabled distribution under H_0. Because $h(\Lambda_n(\mathbf{X}))$ is equivalent to $\Lambda_n(\mathbf{X})$, we specify the size α likelihood ratio test through the test statistic $h(\Lambda_n(\mathbf{X}))$

Some examples.

Examples

11.3 Asymptotic distribution of LR

Theorem 1 (Wilks)**.** *Let $X_1, \ldots, X_n$ be a simple random sample from $f(x, \theta)$, $\theta \in \Theta$, where Θ is an open set of $\mathbb{R}^k$. Consider $H_0 : \theta \in \Theta_0 \leftrightarrow H_1 : \theta \in \Theta - \Theta_0$, where Θ_0 is an open set of $\mathbb{R}^d$ with $d = k - s \geq 1$. Under regularity conditions (Conditions for MLE to be consistent and asymptotic normal) and assuming H_0 is true, then*

$$2log\Lambda_n(\mathbf{X}) \xrightarrow{\mathcal{L}} \chi^2_{k-d} = \chi^2_s, \quad as\ n \to \infty$$

Note: Wilks's theorem depends critically on the fact that not only is Θ open but that Θ_0 is open.

Specification on Θ_0:

- There exists a non-empty subset A in $\mathbb{R}^d, d = k - s \geq 1$ and one-to-one mapping $h = (h_1 \ldots, h_k) : A \mapsto \Theta_0$ where $\partial h / \partial \eta$ is full rank of d at the interior of A, that is,

$$\Theta_0 = \{\theta = h(\eta), \eta \in A\}$$

such that,

$$\{f(x,\theta), \theta \in \Theta_0\} = \{f(x, h(\eta)), \eta \in A\}$$

- Or, Θ_0 can be equivalently expressed (after some transformation) as

$$\Theta_0 = \{\theta = (\theta_1, \ldots, \theta_k) \in \Theta : \theta_i = \theta_{i0}, i = 1, \ldots, s\}$$

  and $(\theta_{10}, \ldots, \theta_{s0}, \theta_{s+1}, \ldots, \theta_{s+k})$ is interior of Θ.

- Or the null can be expressed as s constraints in general:

$$H_0 : g(\theta_0) = 0_{s\times 1}$$

  where $g : \mathbb{R}^k \mapsto \mathbb{R}^s$ and the rank of $\partial g(\theta)/\partial\theta$ is s.

- In a word, $dim\Theta - dim\Theta_0 = s \geq 1$

Proof. Let θ_0 be the true parameter, thus under H_0 we have $\theta_0 = h(\eta_0)$. Write the log likelihood as

$$\ell_n(\theta) = \sum_{i=1}^{n} \log f\left(X_i, \theta\right)$$

and under H_0,

$$\tilde{\ell}_n(\eta) = \sum_{i=1}^{n} \log f\left(X_i, h(\eta)\right)$$

It is easy to see that the Fisher information

$$\tilde{I}(\eta) = BI(\theta)B^T, B = \left(\frac{\partial h}{\partial \eta}\right)_{d\times k}$$

Because the MLE $\widehat{\theta}_n$ satisfies

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \approx I^{-1}(\theta_0)\frac{1}{\sqrt{n}}\nabla_\theta \ell_n(\theta_0) \equiv I^{-1}(\theta_0)U_n(\theta_0),$$

where ∇_θ is the derivative with respect to θ and $U_n(\theta) = \frac{1}{\sqrt{n}}\nabla\ell_n(\theta)$. An expansion of $\ell_n(\theta)$ about $\widehat{\theta}_n$ evaluated at $\theta = \theta_0$ gives

$$\begin{aligned} 2\left[\ell_n(\widehat{\theta}_n) - \ell_n\left(\theta_0\right)\right] &\approx n\left(\widehat{\theta}_n - \theta_0\right)^T I\left(\theta_0\right)\left(\widehat{\theta}_n - \theta_0\right) \\ &\approx U_n(\theta_0)^T I^{-1}(\theta_0) U_n(\theta_0), \end{aligned}$$

Similar argument applied to $\tilde{\ell}_n(\eta)$ to get

$$\sqrt{n}(\widehat{\eta}_n - \eta_0) \approx \tilde{I}^{-1}(\eta_0)\frac{1}{\sqrt{n}}\nabla_\eta\tilde{\ell}_n(\eta_0) \equiv \tilde{I}^{-1}(\eta_0)\tilde{U}_n(\eta_0),$$

and

$$\begin{aligned} 2\left[\tilde{\ell}_n(\widehat{\eta}_n) - \tilde{\ell}_n\left(\eta_0\right)\right] &\approx n\left(\widehat{\eta}_n - \eta_0\right)^T \tilde{I}\left(\eta_0\right)\left(\widehat{\eta}_n - \eta_0\right) \\ &\approx \tilde{U}_n(\eta_0)^T \tilde{I}^{-1}(\eta_0)\tilde{U}_n(\eta_0) \\ &= U_n(\theta_0)^T B^T \tilde{I}^{-1}(\eta_0) B U_n(\theta_0), \end{aligned}$$

where $\tilde{U}_n(\eta) = \frac{1}{\sqrt{n}}\nabla\tilde{\ell}_n(\eta) = BU_n(\theta)$.

Therefor,

$$
\begin{aligned}
2log\Lambda_n(\mathbf{X}) &= 2\left[\ell_n(\widehat{\theta}_n) - \ell_n\left(\theta_0\right)\right] - 2\left[\tilde{\ell}_n(\widehat{\eta}_n) - \tilde{\ell}_n\left(\eta_0\right)\right] \\
&\approx U_n(\theta_0)^T[I^{-1}(\theta_0) - B^T\tilde{I}^{-1}(\eta_0)B]U_n(\theta_0) \\
&= [Z_n]^T[\mathrm{I}_k - I^{1/2}(\theta_0)B^T\tilde{I}^{-1}(\eta_0)BI^{1/2}(\theta_0)][Z_n]
\end{aligned}
$$

where $Z_n = I^{-1/2}(\theta_0)U_n(\theta_0) \sim N(0, \mathrm{I}_k)$.

Denote

$$
G = \mathrm{I}_k - I^{1/2}(\theta_0)B^T\tilde{I}^{-1}(\eta_0)BI^{1/2}(\theta_0)
$$

and note that $\tilde{I}(\eta_0) = BI(\theta_0)B^T$, it is easy to see $G^2 = G$ and thus

$$
Rank(G) = tr(G) = k - tr(\tilde{I}^{-1}(\eta_0)BI(\theta_0)B^T) = k - d = s.
$$

The proof is completed by the property of chi-square distribution. $\square$

11.4 Wald test

- Test on a single parameter $H_0 : \theta = \theta_0$, since the MLE satisfies $\sqrt{n}(\hat{\theta} - \theta) \to N(0, 1/I(\theta))$, the Wald test rejects H_0 when

$$W = \frac{(\hat{\theta} - \theta_0)^2}{Var(\hat{\theta})} = n(\hat{\theta} - \theta_0)i(\hat{\theta})(\hat{\theta} - \theta_0) > \chi_1^2(\alpha).$$

- Test(s) on multiple parameters $H_0 : \theta = \theta_0, \theta \in \mathbb{R}^k$. Since $\sqrt{n}(\widehat{\theta} - \theta) \to N_k(0, I^{-1}(\theta))$, it follows that the Wald test that rejects H_0 when

$$W_n(\theta_0) = n(\widehat{\theta} - \theta_0)^T i(\widehat{\theta})(\widehat{\theta} - \theta_0) \geq \chi_k^2(\alpha).$$

- For test on $H_0 : g(\theta) = 0_{s\times 1}$ the Wald test rejects H_0 when

$$n(g(\hat{\theta}))^T[\nabla g(\hat{\theta}) i^{-1}(\hat{\theta})(\nabla g(\hat{\theta}))^T]^{-1}(g(\hat{\theta})) > \chi_s^2(\alpha)$$

- $i(\hat{\theta})$ usually takes $-\frac{1}{n}\frac{\partial^2 \ell_n}{\partial\theta\partial\theta^T}|_{\theta=\hat{\theta}}$, $I(\theta_0)$ would also do if easily computed.

Example

Find the Wald test for the hypothesis $H_0 : \theta \in \Theta_0 = \{\theta : \theta_j = \theta_{0,j}, j = 1, \ldots, s\}$.

The H_0 can be rewritten as

$$H_0 : g(\theta) = \theta^{(1)} - \theta_0^{(1)} = 0$$

we write the MLE for $\theta \in \Theta$ as $\widehat{\theta}_n = \left(\widehat{\theta}_n^{(1)}, \widehat{\theta}_n^{(2)}\right)$ where $\widehat{\theta}_n^{(1)} = \left(\widehat{\theta}_1, \ldots, \widehat{\theta}_s\right)$ and $\widehat{\theta}_n^{(2)} = \left(\widehat{\theta}_{s+1}, \ldots, \hat{\theta}_k\right)$ and define the Wald statistic as

$$W_n\left(\theta_0^{(1)}\right) = n\left(\widehat{\theta}_n^{(1)} - \theta_0^{(1)}\right)^T \left[I^{11}\left(\widehat{\theta}_n\right)\right]^{-1} \left(\widehat{\theta}_n^{(1)} - \theta_0^{(1)}\right) \xrightarrow{\mathcal{L}} \chi_s^2, \text{under} H_0.$$

where $I^{11}(\theta)$ is the upper diagonal block of $I^{-1}(\theta)$ written as

$$I^{-1}(\theta) = \begin{pmatrix} I^{11}(\theta) & I^{12}(\theta) \\ I^{21}(\theta) & I^{22}(\theta) \end{pmatrix}$$

11.5 Score test

- For the simple hypothesis $H_0 : \theta = \theta_0$, Rao's score test or Lagrange Multipliers test is based on the observation that, letting H be the Lagrangian:

$$H = \ell_n(\theta) - \lambda' \left(\theta - \theta^0\right),$$

the first-order conditions are:

$$\frac{\partial \ell_n}{\partial \theta} = \lambda; \quad \theta = \theta^0,$$

so $\lambda/\sqrt{n} = U_n(\theta_0) \xrightarrow{\mathcal{L}} N\left(0, I(\theta_0)\right)$. Thus, $LM_n = \lambda^T I^{-1}(\theta_0)\lambda/n \xrightarrow{\mathcal{L}} \chi^2_k$ under H_0.

- Equivalently, it follows from this that under H_0, as $n \to \infty$,

$$LM_n\left(\theta_0\right) = U_n^T\left(\theta_0\right) I^{-1}\left(\theta_0\right) U_n\left(\theta_0\right) \xrightarrow{\mathcal{L}} \chi^2_k.$$

The test rejects H_0 when $LM_n\left(\theta_0\right) \geq \chi^2_k(\alpha)$.

- For test on $H_0 : g(\theta) = 0_{s\times 1}$, the statistic employed in the score test is based on the ML estimate $\widehat{\theta}_n^R$ that is obtained from the solution of the constrained optimization problem

$$\widehat{\theta}_n^R = \arg\max_{\theta \in H_0} \ell_n(\theta)$$

The test statistic, called score statistic, is

$$LM_n = U_n(\hat{\theta}_n^R)^\top I^{-1}(\widehat{\theta}_n^R) U_n(\widehat{\theta}_n^R) \xrightarrow{\mathcal{L}} \chi_s^2, \ under \ H_0$$

- Note that $I^{-1}(\theta_0)$ depends on the unknown parameter value θ_0. We can evaluate the terms involved in θ_0 at the constrained MLE, $\widehat{\theta}_n^R$ to get a usable statistic. We can approximate $I(\theta_0)$ with either $-\frac{1}{n}\nabla^2 \ell_n(\widehat{\theta}_n^R)$ or $\frac{1}{n}\nabla \ell_n(\widehat{\theta}_n^R)\nabla^\top \ell_n(\widehat{\theta}_n^R)$.

Proof

Proof. FOC's are given by

$$\begin{aligned} \nabla \ell_n(\widehat{\theta}_n^R) + \nabla g(\widehat{\theta}_n^R)\lambda &= 0 \\ g(\widehat{\theta}_n^R) &= 0 \end{aligned}$$

First order Taylor expansions of $g(\widehat{\theta}_n)$ and $g(\widehat{\theta}_n^R)$ around θ_0 gives, ignoring $o_p(1)$ terms,

$$\begin{aligned} \sqrt{n}g(\widehat{\theta}_n) &= \sqrt{n}g(\theta_0) + \nabla g^\top(\theta_0)\sqrt{n}(\widehat{\theta}_n - \theta_0) \\ \sqrt{n}g(\widehat{\theta}_n^R) &= \sqrt{n}g(\theta_0) + \nabla g^\top(\theta_0)\sqrt{n}(\widehat{\theta}_n^R - \theta_0) \end{aligned}$$

Note that $g(\widehat{\theta}_n^R) = 0$, we have

$$\sqrt{n}g(\widehat{\theta}_n) = \nabla g^\top(\theta_0)\sqrt{n}(\widehat{\theta}_n - \widehat{\theta}_n^R)$$

On the other hand, taking first order Taylor series expansions of $\nabla \ell_n(\widehat{\theta}_n)$ and $\nabla \ell_n(\widehat{\theta}_n^R)$ around θ_0 gives, ignoring $o_p(1)$ terms,

$$0 = \nabla \ell_n(\widehat{\theta}_n) = \nabla \ell_n(\theta_0) + \nabla^2 \ell_n(\theta_0)(\widehat{\theta}_n - \theta_0) \Rightarrow$$

$$0 = U_n(\widehat{\theta}_n) = U_n(\theta_0) - I(\theta_0)\sqrt{n}(\widehat{\theta}_n - \theta_0),$$

note that $-\nabla^2 \ell_n(\theta_0) \to I(\theta_0)$ by the law of large numbers. Similarly,

$$U_n(\widehat{\theta}_n^R) = \nabla \ell_n(\widehat{\theta}_n^R)/\sqrt{n} = U_n(\theta_0) - I(\theta_0)\sqrt{n}(\widehat{\theta}_n^R - \theta_0).$$

Taking the difference, we have

$$U_n(\widehat{\theta}_n^R) = I(\theta_0)\sqrt{n}(\widehat{\theta}_n - \widehat{\theta}_n^R).$$

Hence,

$$\sqrt{n}(\widehat{\theta}_n - \widehat{\theta}_n^R) = I^{-1}(\theta_0)U_n(\widehat{\theta}_n^R).$$

Therefore,

$$\sqrt{n}g(\widehat{\theta}_n) = \nabla g^\top(\theta_0)I^{-1}(\theta_0)U_n(\widehat{\theta}_n^R).$$

We then deduce

$$\begin{aligned}\sqrt{n}g(\widehat{\theta}_n) &= -\nabla g^\top(\theta_0)I^{-1}(\theta_0)\nabla g(\widehat{\theta}_n^R)\frac{\lambda}{\sqrt{n}} \\ &\to -\nabla g^\top(\theta_0)I^{-1}(\theta_0)\nabla g(\theta_0)\frac{\lambda}{\sqrt{n}},\end{aligned}$$

since $\widehat{\theta}_n^R \to \theta_0$ hence $g(\widehat{\theta}_n^R) \to g(\theta_0)$ under the null hypothesis. By $\sqrt{n} g(\widehat{\theta}_n) \xrightarrow{\mathcal{L}} N(0, \nabla g^\top(\theta_0) I^{-1}(\theta_0) \nabla g(\theta_0))$ under H_0, we have

$$\frac{\lambda}{\sqrt{n}} \xrightarrow{\mathcal{L}} N\left(0, \left(\nabla g^\top(\theta_0) I^{-1}(\theta_0) \nabla g(\theta_0)\right)^{-1}\right) \quad \text{under } H_0$$

Again, forming the quadratic form of the normal random variables, we obtain

$$\frac{1}{n} \lambda^\top \left(\nabla g^\top(\theta_0) I^{-1}(\theta_0) \nabla g(\theta_0)\right) \lambda \xrightarrow{\mathcal{L}} \chi_s^2 \quad \text{under } H_0$$

Alternatively, using $\nabla g(\theta_0) \frac{\lambda}{\sqrt{n}} \approx U_n(\widehat{\theta}_n^R)$, another form of the test statistic is given by

$$U_n^\top(\widehat{\theta}_n^R) I^{-1}(\widehat{\theta}_n^R) U_n(\widehat{\theta}_n^R) \xrightarrow{\mathcal{L}} \chi_s^2 \quad \text{under } H_0$$

$\square$

Example

Find the Score test for the hypothesis $H_0 : \theta \in \Theta_0 = \{\theta : \theta_j = \theta_{0,j}, j = 1, \ldots, s\}$.

The H_0 can be rewritten as

$$H_0 : g(\theta) = \theta^{(1)} - \theta_0^{(1)} = 0$$

$\theta_0^{(1)} = (\theta_{0,1}, \ldots, \theta_{0,s})$. $\theta_0 = (\theta_0^{(1)}, \theta_0^{(2)})$. We write the restricted MLE for $\theta \in \Theta$ as $\widehat{\theta}_n^R = (\theta_0^{(1)}, \widehat{\theta}_n^{(2)})$ where $\widehat{\theta}_n^{(2)} = (\widehat{\theta}_{s+1}, \ldots, \hat{\theta}_k)$ and define the Score statistic as

$$LM = \frac{1}{n}\lambda^\top \left(\nabla g'(\theta_0) I^{-1}(\theta_0) \nabla g(\theta_0)\right) \lambda = \frac{1}{n}\lambda^\top I^{11}(\theta_0)\lambda$$

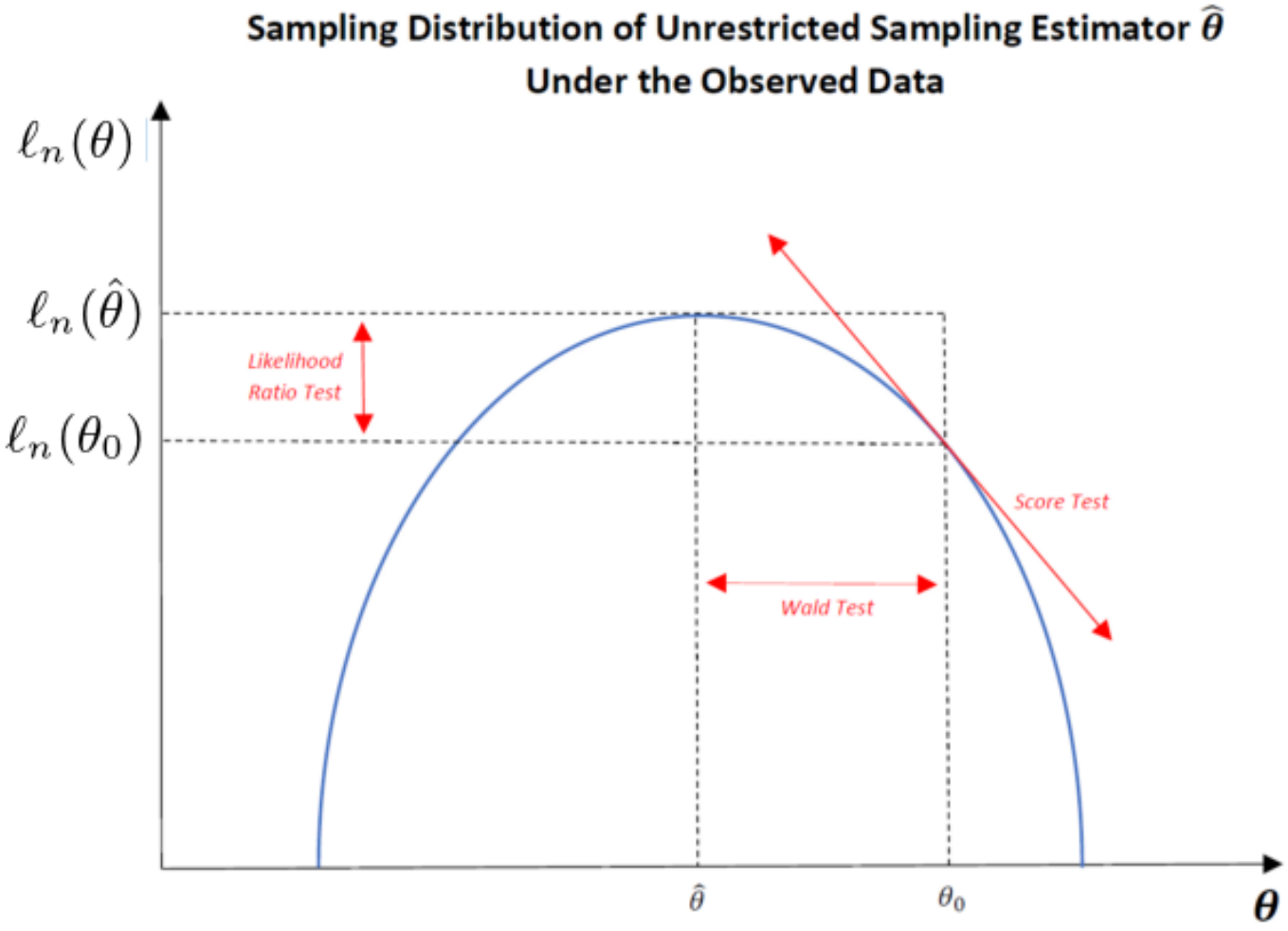
Alternatively, another form of the test statistic is given by

$$LM = U_n^\top(\widehat{\theta}_n^R) I^{-1}(\theta_0) U_n(\widehat{\theta}_n^R) = \frac{1}{n}\frac{\partial \ell_n(\widehat{\theta}_n^R)}{\partial \theta'^{(1)}} I^{11}(\theta_0) \frac{\partial \ell_n(\widehat{\theta}_n^R)}{\partial \theta^{(1)}}$$

Summary

All of the test statistics mentioned here are asymptotically equivalent to the LRT (Wald and Rao score alike).

- an LRT requires computation of an MLE, $\widehat{\theta}_n$, and a restricted MLE, $\widehat{\theta}_n^R$, and it is a function of the distance between $\ell_n(\hat{\theta}_n)$ and $\ell_n(\widehat{\theta}_n^R)$
- a Wald test requires only computation of the MLE $\widehat{\theta}_n$, and it is a function of the distance between $g(\hat{\theta}_n)$ and $g(\theta_0) = 0$.
- a Rao score test requires only computation of the restricted MLE $\widehat{\theta}_n^R$ and measures how far the Lagrangian multiplier is from zero.
- As shown above, Wald, LM and LR test are asymptotically equivalent with χ_s^2.
- LRT is usually more powerful than Wald and Rao score tests under small or moderate sample size.



LRT, Wald and Score tests for $H_0 : \theta = \theta_0$

11.6 Confidence Regions

- Any of the tests discussed above (LRT, Wald, score) can be inverted to find confidence regions for the values of l parametric functions

$$\mathbf{u} = (u_1(\theta), u_2(\theta), \ldots, u_l(\theta))'$$

- That is, for any vector of potential values for these functions $\boldsymbol{c} = (c_1, c_2, \ldots, c_l)'$, one may define $g_{\boldsymbol{c},i}(\theta) = u_i(\theta) - c_i$ and a test of some type above for

$$\mathrm{H}_0 : \boldsymbol{g}_{\boldsymbol{c}}(\theta) = \mathbf{u} - \mathbf{c} = \mathbf{0}$$

  The set of all $\boldsymbol{c}$ for which an approximately α-level test does not reject constitutes an approximately $(1-\alpha) \times 100\%$ confidence set for the vector $(u_1(\theta), u_2(\theta), \ldots, u_l(\theta))'$.

- When Wald tests are used,the regions will be ellipsoidal:

$$CR = \{\mathbf{c} : [\mathbf{c} - \mathbf{u}(\widehat{\theta})]' Var^{-1}(\mathbf{u}(\hat{\theta}))[\mathbf{c} - \mathbf{u}(\widehat{\theta})] \leq \chi^2_l(\alpha)\}$$